



Project Proposal

Course Title : Database Management System Lab

Project Title : Custom PC Builder

Course Code : CSE-224

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1. Introduction

A Custom PC Builder is a web-based application that allows users to create personalized PC configurations by selecting components from a predefined catalog. The system uses a frontend for user interaction, a PHP backend for processing requests and applying basic compatibility logic, and a MySQL database for storing component information and saved PC builds. The application is designed as a PC build planning tool rather than an e-commerce or inventory management system.

2. Problem Statement

Planning a custom PC build manually can make it difficult to organize selected components, compare different build configurations, calculate the total cost, and avoid basic incompatibility between the CPU and motherboard. A web-based PC Builder provides a structured way to select components, calculate the total price, perform a basic socket compatibility check, and save, view, update, or delete PC builds.

3. Objectives

- Develop a complete web application demonstrating CRUD operations.
- Design a clear and suitable MySQL relational database.
- Connect the frontend, backend, and database.
- Allow users to create and manage custom PC builds.
- Automatically calculate the total price of selected components.
- Implement a basic CPU–Motherboard socket compatibility check.
- Provide a simple and user-friendly interface.

4. Technologies

Frontend: HTML, CSS, JavaScript

Backend: PHP

Database: MySQL

Server: XAMPP

5. Modules

- **Component Catalog:** Displays a small predefined collection of components by category.
- **PC Build Creator:** Allows users to select components for a custom PC build.
- **Compatibility Check:** Verifies that the selected CPU socket matches the selected motherboard socket.
- **Build Summary:** Displays selected components and automatically calculates the total price.
- **Saved Builds Management:** Allows users to save, view, edit, and delete PC builds.

6. Database Tables

The proposed database uses separate tables for the seven fixed component categories because different hardware categories require different attributes. Each category will contain no more than 10 predefined components. The main tables are: CPUs, Motherboards, GPUs, RAM, Storage, PSUs, Cases, and Builds.

7. CRUD Operations

Create : Create and save a new PC build.

Read : View saved PC builds and read predefined component data.

Update : Modify an existing build by replacing or changing selected components.

Delete : Remove a saved PC build.

The predefined component catalog is primarily read-only. Full CRUD operations are demonstrated through the management of saved PC builds.

8. Proposed Database Design

The proposed database contains seven component tables and one main Builds table. The Builds table stores foreign keys referencing the selected component from each category.

Table	Important Fields
CPUs	CPU_ID (PK), Name, Brand, Socket, Price
Motherboards	Motherboard_ID (PK), Name, Brand, Socket, Price
GPUs	GPU_ID (PK), Name, Brand, Price
RAM	RAM_ID (PK), Name, Brand, Capacity, Speed, Price
Storage	Storage_ID (PK), Name, Brand, Capacity, Price
PSUs	PSU_ID (PK), Name, Brand, Wattage, Price
Cases	Case_ID (PK), Name, Brand, Price
Builds	Build_ID (PK), Build_Name, CPU_ID (FK), Motherboard_ID (FK), GPU_ID (FK), RAM_ID (FK), Storage_ID (FK), PSU_ID (FK), Case_ID (FK), Total_Price

Design assumptions:

All motherboards use the ATX form factor, all cases support ATX motherboards, all RAM options are DDR4, all PSUs provide sufficient wattage for the available combinations, and other advanced compatibility constraints are outside the project scope. The only enforced compatibility rule is CPU socket matching with motherboard socket.

9. Functional Requirements

Users should be able to:

- Browse predefined components by category.
- Select one component from each category.
- Receive basic CPU–Motherboard socket compatibility validation.
- View selected components and the automatically calculated total price.
- Save a new PC build.
- View saved builds.
- Edit an existing build.
- Delete a saved build.

10. Non-Functional Requirements

- Simple and user-friendly interface.
- Reliable connection between the PHP backend and MySQL database.
- Responsive layout for common screen sizes.
- Consistent and accurate data handling.
- Reasonable performance for the small predefined component dataset.

11. Expected Output

A fully functional web application connected to a MySQL database where users can create custom PC configurations, view selected components, receive basic CPU–Motherboard compatibility feedback, calculate total build price, and perform Create, Read, Update, and Delete operations on saved PC builds.

12. Timeline

Week 1: Requirement Analysis

Week 2: Database Design (ER Diagram)

Week 3: Frontend Development

Week 4: Backend Development

Week 5: Database Integration & CRUD **Week 6:** Testing & Final Report

13. Conclusion

This project demonstrates practical knowledge of frontend development, backend programming, MySQL database design, SQL queries, relationships, and CRUD operations through the development of a Custom PC Builder. The project maintains a manageable scope by using a small predefined component catalog and a single basic compatibility rule while still demonstrating complete frontend–backend–database integration.